

George Joshi

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Education

University of Texas at Arlington

MS, Computer Science

Expected Dec 2027

Tribhuvan University

BS in Computer Engineering

July 2024

Publications

- **A Comparative Study on the Effect of News Sentiment on Stock Price Prediction with Deep Learning**
Conducted research integrating news sentiment with historical stock data using **LSTM** and **GRU** models. Evaluation with **RMSE**, **MAPE**, and correlation coefficient (**R**) demonstrated improved predictive accuracy.

Experience

Research Assistant

ACL2 Lab, University of Texas at Arlington

Arlington, Texas Jan

2026 – Present

- Researching **computational bioacoustics** and cross-species birdsong tokenization; evaluating codebook-based acoustic representations (shape, CV, V, k-means, random) across **mel-spectrogram** and **BirdAVES embedding** spaces.
- Found species-derived **shape tokens** reconstruct near-ceiling while human-phonetic codebooks fall at or below the noise floor, a key negative result on cross-domain phonetic transfer; prepared for **ACL Rolling Review (EMNLP)** submission.
- Built and scaled an audio processing pipeline for a **110K+ file** birdsong dataset on a multi-GPU research cluster, resolving environment and dependency conflicts across Python versions.
- Developed a syllable-to-phone transcription system using **MFCC** feature extraction and **kNN** classification, improving label stability with clustering-based equivalence classes.

AI Engineer

DL.surf

Bhaktapur, Kathmandu

Aug 2024 – July 2025

- **Built** a semantic **Search & Re-ranking system** (**150K+** daily **FastAPI** requests) using **all-MiniLM-L6-v2**, **Milvus**, and **Cohere Reranker**; reduced query latency by **~60%** with **Redis caching** and **DB stored procedures**; maintained **CI/CD pipelines**.
- **Developed** a **low-latency, near real-time NSFW classifier** for images, videos, and text; optimized models (**PyTorch** → **ONNX** → **quantized TFLite**) and integrated with **file upload/commenting APIs**; performed **integration & latency testing** to ensure scalability.
- Developed **Grafana dashboards** for real-time monitoring: one with **Prometheus + Grafana** for system performance, and another with **Postgres + Grafana** for tracking platform **User Metrics**.
- Configured and optimized **PostgreSQL** with working memory limits, connection pooling, and indexing to improve query performance and system reliability.
- Implemented **Role-Based Access Control (RBAC)** in PostgreSQL to enforce fine-grained file access permissions aligned with user roles.
- Designed and maintained **stored procedures, query optimizations, and backup/recovery strategies** to ensure high availability and data integrity.

Machine Learning Intern

Treeleaf Technologies

Kathmandu, Nepal

May 2023 – Jul 2023

- Developed and optimized **YOLOv7**-based computer vision models for object detection.
- Collaborated on the development and deployment of machine learning models.
- Built automated pipelines for deploying custom ML models to production.
- Implemented **Git**-based version control to facilitate team collaboration.

Projects & Simulations

British Airways Data Science Job Simulation

- Scraped and analyzed customer review data to derive actionable insights.
- Performed feature analysis to identify key factors influencing customer buying behavior.

Cognizant Artificial Intelligence Job Simulation

- Conducted exploratory data analysis (EDA) using Python and Jupyter for Gala Groceries.

- Developed a reusable Python module for model training, evaluation, and validation.
- Delivered business insights through a comprehensive PowerPoint presentation.

Technologies

Programming Languages: Python, SQL, C++, JavaScript

Data Visualization: Matplotlib, Tableau, Seaborn, Streamlit, Plotly

DevOps & Cloud: GitHub Actions, AWS (EC2, ECR, S3), Docker, Jenkins, Kubernetes, Prometheus, Grafana

Data Science Frameworks: TensorFlow, PyTorch, OpenCV, scikit-learn, YOLO, Keras, Statsmodels, HuggingFace, MLflow

Data Engineering & Databases: PostgreSQL, Milvus, Kafka, MongoDB, Excel

Certifications

Foundations of Data Science – Stanford Online

Introduction to Statistics – Stanford Online

Deep Learning Course – Scaler

Statistical Methods for Decision Making – Scaler

OpenCV Bootcamp – Great Learning